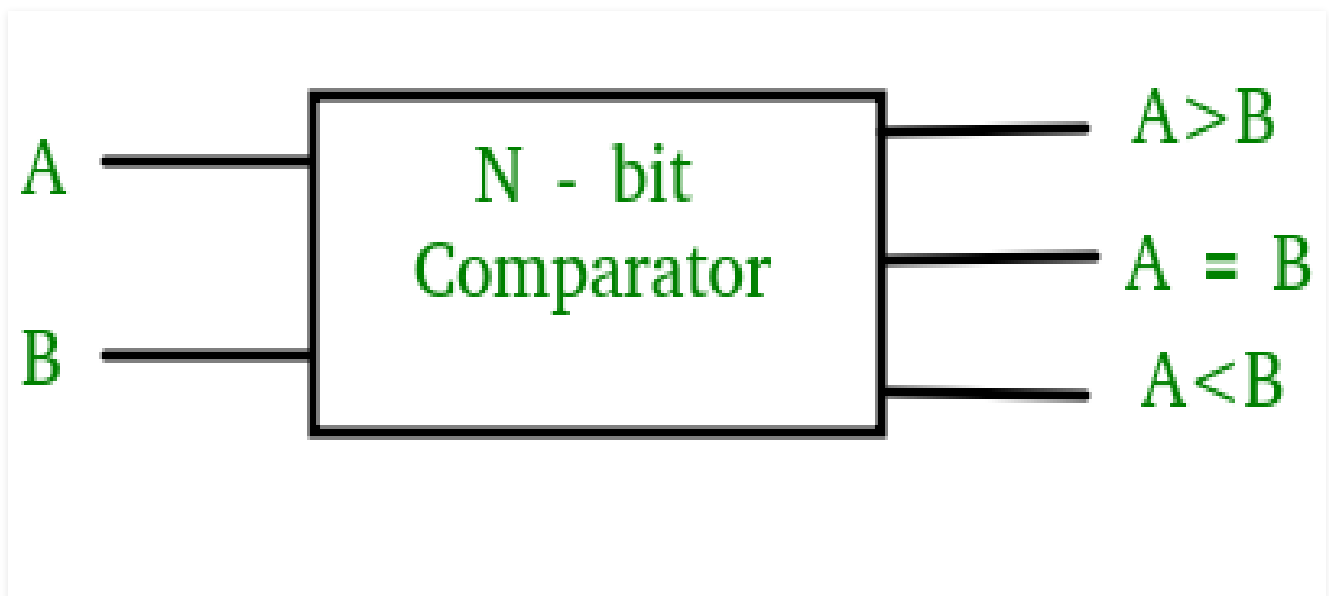


Magnitude Comparator in Digital Logic

Difficulty Level : Easy • Last Updated : 19 Feb, 2021

A magnitude digital Comparator is a combinational circuit that **compares two digital or binary numbers** in order to find out whether one binary number is equal, less than or greater than the other binary number. We logically design a circuit for which we will have two inputs one for A and other for B and have three output terminals, one for $A > B$ condition, one for $A = B$ condition and one for $A < B$ condition.

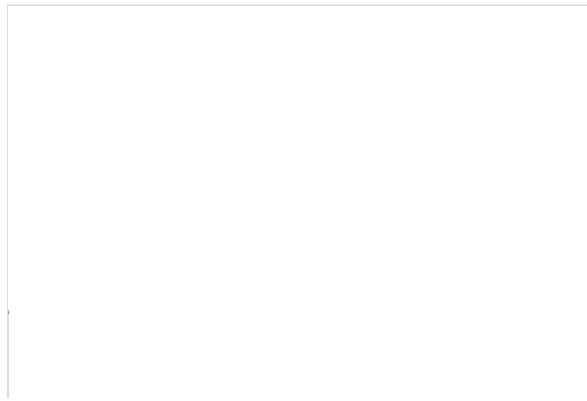


1-Bit Magnitude Comparator –

A comparator used to compare two bits is called a single bit comparator. It consists of two inputs each for two single bit numbers and three outputs to generate less than

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The truth table for a 1-bit comparator is given below:

A	B	$A < B$	$A = B$	$A > B$
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

From the above truth table logical expressions for each output can be expressed as follows:

$$A > B: AB'$$

$$A < B: A'B$$

$$A = B: A'B' + AB$$

From the above expressions we can derive the following formula:

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$$(A < B) + (A > B) = A'B + AB'$$

Taking complement both sides

$$(A < B) + (A > B)' = (A'B + AB')'$$

$$(A < B) + (A > B)' = (A'B)' (AB')'$$

$$(A < B) + (A > B)' = (A + B') (A' + B)$$

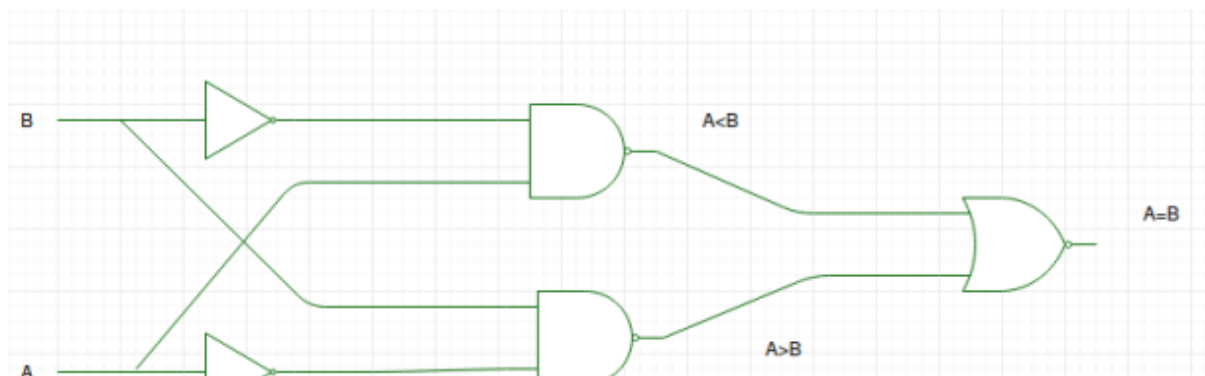
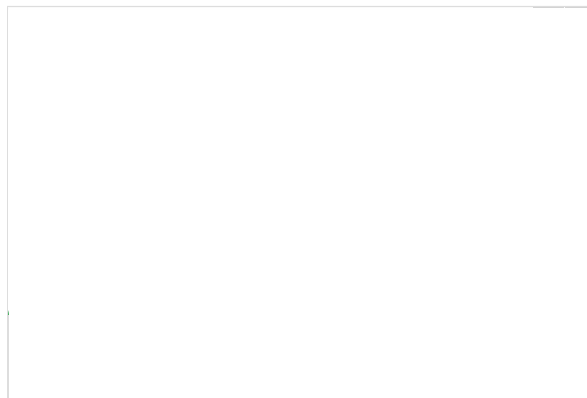
$$(A < B) + (A > B)' = (AA' + AB + A'B' + BB')$$

$$= (AB + A'B')$$

Thus,

$$(A < B) + (A > B)' = (A = B)$$

By using these Boolean expressions, we can implement a logic circuit for this comparator as given below:



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2-Bit Magnitude Comparator –

A comparator used to compare two binary numbers each of two bits is called a 2-bit Magnitude comparator. It consists of four inputs and three outputs to generate less than, equal to and greater than between two binary numbers.

The truth table for a 2-bit comparator is given below:

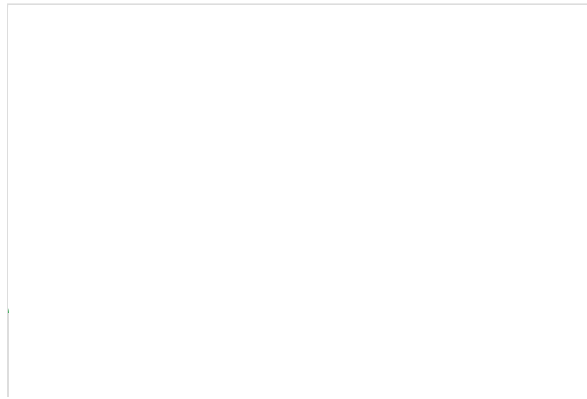
INPUT				OUTPUT		
A1	A0	B1	B0	A<B	A=B	A>B
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

From the above truth table K-map for each output can be drawn as follows:

		B1B0					
		00	01	11	10		
A1A0	00	0	0	0	0		
	01	1	0	0	0		
	11	1	1	0	1		
	10	1	1	0	0		

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		A = B			
		B1B0	00	01	11
A1A0	00	1	0	0	0
	01	0	1	0	0
	11	0	0	1	0
	10	0	0	0	1

		A < B			
		B1B0	00	01	11
A1A0	00	0	1	1	1
	01	0	0	1	1
	11	0	0	0	0
	10	0	0	0	0

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From the above K-maps logical expressions for each output can be expressed as follows:

$$A > B: A_1 B_1' + A_0 B_1' B_0' + A_1 A_0 B_0'$$

$$A = B: A_1' A_0' B_1' B_0' + A_1' A_0 B_1' B_0 + A_1 A_0 B_1 B_0 + A_1 A_0' B_1 B_0'$$

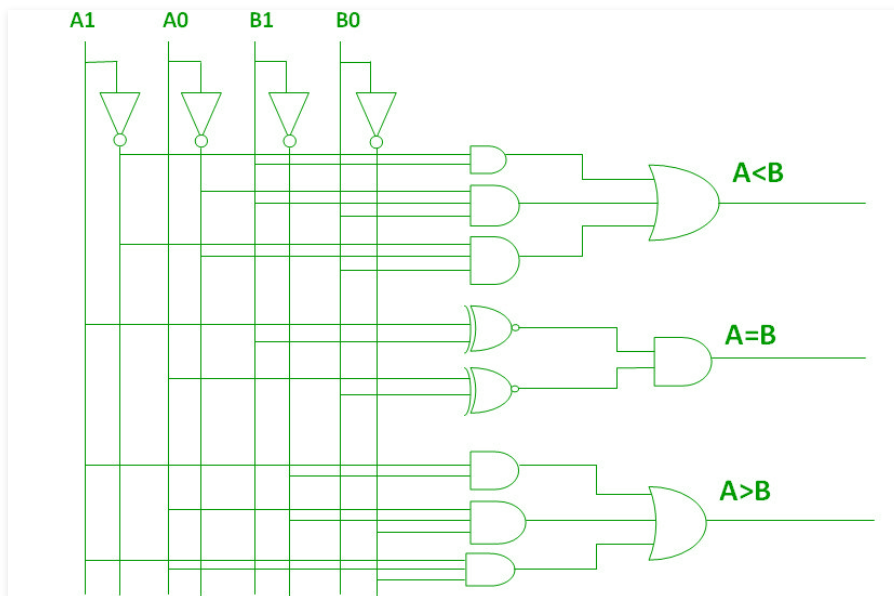
$$: A_1' B_1' (A_0' B_0' + A_0 B_0) + A_1 B_1 (A_0 B_0 + A_0' B_0')$$

$$: (A_0 B_0 + A_0' B_0') (A_1 B_1 + A_1' B_1')$$

$$: (A_0 \text{ Ex-Nor } B_0) (A_1 \text{ Ex-Nor } B_1)$$

$$A < B: A_1' B_1 + A_0' B_1 B_0 + A_1' A_0' B_0$$

By using these Boolean expressions, we can implement a logic circuit for this comparator as given below:



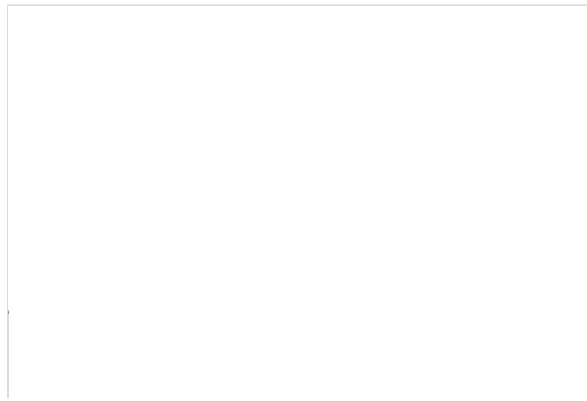
4-Bit Magnitude Comparator –

A comparator used to compare two binary numbers each of four bits is called a 4-bit magnitude comparator. It consists of eight inputs each for two four bit numbers and three outputs to generate less than, equal to and greater than between two binary numbers.

In a 4-bit comparator the condition of $A > B$ can be possible in the following four cases:

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1. If $A_3 = 1$ and $B_3 = 0$
2. If $A_3 = B_3$ and $A_2 = 1$ and $B_2 = 0$
3. If $A_3 = B_3$, $A_2 = B_2$ and $A_1 = 1$ and $B_1 = 0$
4. If $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$ and $A_0 = 1$ and $B_0 = 0$

Similarly the condition for $A < B$ can be possible in the following four cases:

1. If $A_3 = 0$ and $B_3 = 1$
2. If $A_3 = B_3$ and $A_2 = 0$ and $B_2 = 1$
3. If $A_3 = B_3$, $A_2 = B_2$ and $A_1 = 0$ and $B_1 = 1$
4. If $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$ and $A_0 = 0$ and $B_0 = 1$

The condition of $A=B$ is possible only when all the individual bits of one number exactly coincide with corresponding bits of another number.

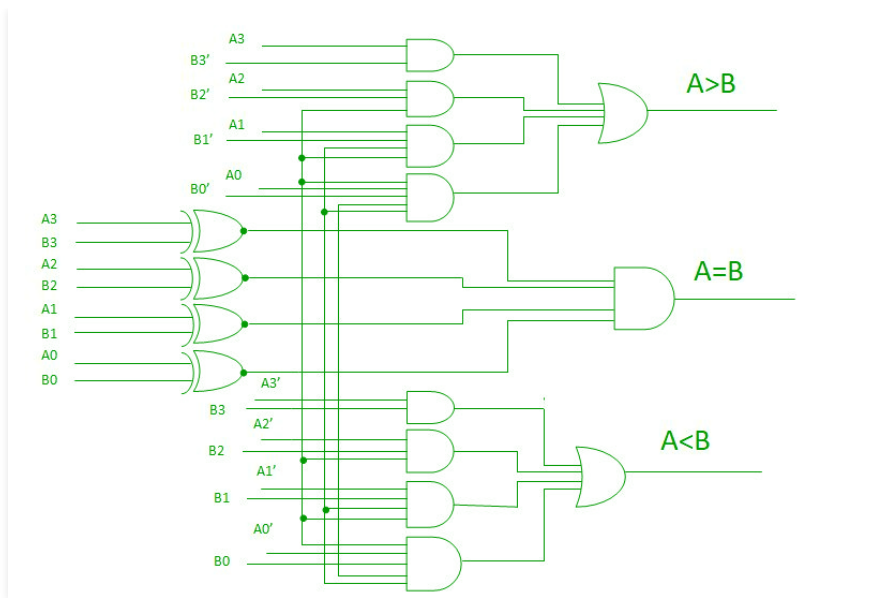
From the above statements logical expressions for each output can be expressed as follows:

$A < B: (A_3 \text{ Ex-Nor } B_3) (A_2 \text{ Ex-Nor } B_2) (A_1 \text{ Ex-Nor } B_1) (A_0 \text{ Ex-Nor } B_0)$

$A > B: (A_3 \text{ Ex-Nor } B_3) (A_2 \text{ Ex-Nor } B_2) (A_1 \text{ Ex-Nor } B_1) (A_0 \text{ Ex-Nor } B_0)$

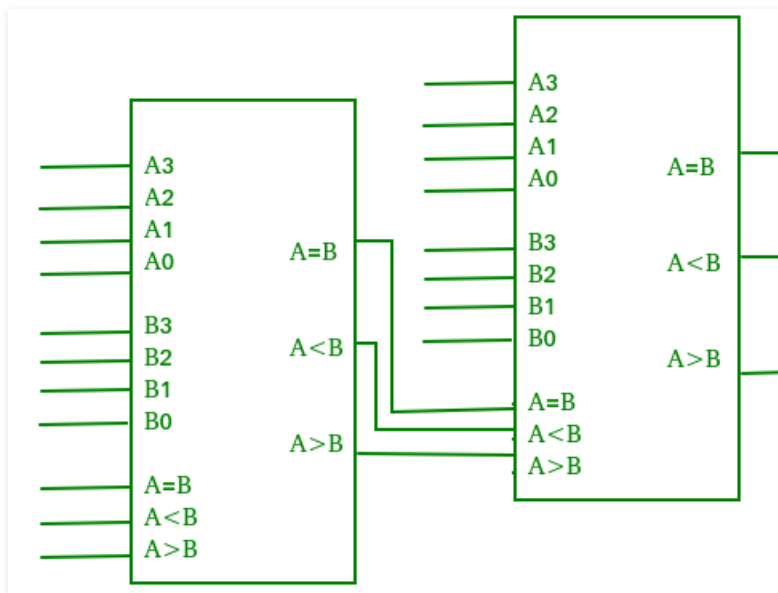
$A = B: (A_3 \text{ Ex-Nor } B_3) (A_2 \text{ Ex-Nor } B_2) (A_1 \text{ Ex-Nor } B_1) (A_0 \text{ Ex-Nor } B_0)$

By using these Boolean expressions, we can implement a logic circuit for this comparator as given below:



Cascading Comparator –

A comparator performing the comparison operation to more than four bits by cascading two or more 4-bit comparators is called cascading comparator. When two comparators are to be cascaded, the outputs of the lower-order comparator are connected to corresponding inputs of the higher-order comparator.



1. Comparators are used in central processing units (CPUs) and microcontrollers (MCUs).
2. These are used in control applications in which the binary numbers representing physical variables such as temperature, position, etc. are compared with a reference value.
3. Comparators are also used as process controllers and for Servo motor control.
4. Used in password verification and biometric applications.

References –

[Digital comparator – Wikipedia](#)

[Comparator – epgp.inflibnet](#)

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